

# Task #267 Cosmological cycle observational signatures v0.2 — candidate observable enumeration

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## Task #267 Cosmological Cycle Observational Signatures v0.2

v0.1 → v0.2 extension

v0.1 (INV-4874) filed opening structural scope. v0.2 enumerates candidate observable signatures at categorical level for multi-month research targeting.

BST anchor theorems for cosmological cycle

| Anchor     | Statement                                                                               | Observable implication                                              |
|------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| T2417      | Substrate Cognition Cosmological Extension v0.2<br>(DOUBLE-LOCKED EXTERNAL per Cal #50) | Substrate-cycle period scaling vs cosmological age $T_0$            |
| T2418      | $\Lambda \leftrightarrow$ Casimir vacuum unification                                    | Time-evolving $\Lambda(t)$ from substrate-Zone-2 vacuum dynamics    |
| T2420      | Four-Zone Vacuum Decomposition                                                          | Per-Zone vacuum-energy budget; Zone-2 substrate contribution to CMB |
| T2405      | Koons tick $t_{\text{Planck}} \cdot \alpha^{(C_2^2)} \approx 10^{-120} \text{ s}$       | Sub-Planck substrate clock; cosmological cycle period anchors       |
| T2469 SCMP | Substrate moderates observer-recording coherence (Layer 2 INTERNAL)                     | Bell sub-Tsirelson at $1/2^{N_c}$ — Layer 1 EXTERNAL-eligible       |

## Candidate observable signatures (v0.2 enumeration)

Category A — CMB Power Spectrum

A1. CMB high-multipole residuals - BST predicts substrate-cycle imprint on CMB power spectrum at high- $l$  where cosmological-cycle period aliases into observable angular scales - Comparison:  $\Lambda$ CDM null hypothesis vs BST substrate-cycle modulation - Datasets: Planck high- $l$ , ACT, SPT - Multi-month: requires substrate-cycle period prediction from T2405 Koons tick +  $N_{\text{max}}$  scaling

A2. CMB B-mode polarization at substrate-cycle resonance - Possibility: cosmological-cycle phase imprint on B-mode polarization at specific multipoles - BST framework: substrate-Zone-2 vacuum oscillations couple to CMB photon polarization via  $\Lambda \leftrightarrow$  Casimir - Datasets: BICEP/Keck, LiteBIRD, CMB-S4 (future)

## Category B — Dark Energy $w(z)$ Reconstruction

B1. DESI  $w_0 \neq -1$  watching brief (current BST registry) - BST prediction: substrate-cycle yields time-evolving  $w(z)$  signature - Current observation: DESI 2024 finds  $w_0 = -0.65 \pm 0.16$  ( $3\sigma$  tension with  $\Lambda$ CDM) - BST tier: I-tier (compatible if substrate-cycle  $\Lambda$ -evolution predicted matches) - Datasets: DESI, Euclid, LSST

B2. Dark energy equation-of-state oscillation - BST: oscillatory  $w(z)$  from substrate-cycle 4-Zone commitment dynamics (T2420) - Signature:  $w(z)$  shows substrate-cycle phase coherence at multi-Gyr period - Multi-month: prediction-from-substrate-cycle-period derivation pending

## Category C — Large-Scale Structure

C1. BAO scale evolution - BAO acoustic scale  $r_d \sim 147$  Mpc; BST may predict substrate-cycle imprint on BAO drift - Datasets: DESI, Euclid, eBOSS

C2. ISW (Integrated Sachs-Wolfe) effect anomaly - BST: time-evolving  $\Lambda$  produces enhanced/suppressed ISW signal - Cross-correlate CMB with LSST galaxy distribution

C3. Void scaling law substrate signature - BST: substrate-cycle phase coherence in cosmic void size distribution - Anchor: void number density-radius scaling; substrate-cycle modulation hypothesis - Datasets: SDSS, DESI void catalogs

## Category D — Substrate-Period Identification

D1. Cosmological cycle period from substrate parameters - Candidate:  $T_{\text{cycle}} = T_0 \cdot \alpha^k$  for some BST-natural exponent  $k$  - Or:  $T_{\text{cycle}} = t_{\text{Planck}} \cdot \alpha^{(C_2^2)} \cdot M^j$  for substrate-Mersenne exponent  $j$  - Anchor: T2405 Koons tick ( $10^{-120}$  s) +  $N_{\text{max}}$  scaling

D2. Cosmological  $Z_2$  reflection signature (Casey W-21 parity) - Per T2472 (Parity  $P_{\text{op}}$ ): substrate Möbius involution may have cosmological-scale  $Z_2$  reflection - Observable: parity-violating signatures in large-scale structure (chirality at cosmological scales)

## Category E — Cross-cosmology comparison

E1. CCC (Conformal Cyclic Cosmology — Penrose) discrimination - BST cosmological cycle vs CCC: different period predictions + different observable signatures - BST distinguishes via substrate-tick anchor (T2405) vs Penrose conformal eon transition - Outreach: Penrose contact previously identified (project\_outreach\_contacts.md)

E2. LQC (Loop Quantum Cosmology) bounce vs BST cycle - LQC predicts pre-Big-Bang bounce; BST cosmological cycle predicts oscillatory substrate-period - Observable discriminator: high- $l$  CMB residuals + B-mode polarization at substrate-period resonance

## Multi-month checkpoint cadence

Per Keeper “no HOLDS” directive: workable multi-month items stay on board with sub-month checkpoint cadence.

| Checkpoint  | Target                 | Deliverable                                                              |
|-------------|------------------------|--------------------------------------------------------------------------|
| v0.2 (this) | Friday May 22, 2026    | Observable candidate enumeration                                         |
| v0.3        | Sub-month (~June 2026) | Per-category BST<br>observable-prediction derivation                     |
| v0.4        | Multi-month            | Quantitative falsifier specification<br>per observable                   |
| v1.0        | TBD multi-month        | Full T2417 + T2418 cosmological<br>framework with explicit falsifier set |

## DOUBLE-LOCKED EXTERNAL register discipline (Cal #50)

External-facing artifacts (papers, outreach letters, Zenodo cover-notes) use: - “BST identifies cosmological cycle signature in [observable]” - “BST predicts time-evolving  $\Lambda$  via T2418  $\Lambda \leftrightarrow$  Casimir vacuum unification” - “Bell sub-Tsirelson 1/8 falsifier at SP-30 Bell experiment design”

NOT: - “Substrate consciousness cycles cosmologically” - “Cosmological cycle = universe thinking” - Substrate-cognition framing in any external register

## Cross-link to SP-30 Substrate Engineering Program

Task #267 cosmological-cycle observable scan provides experimental falsifier targets for SP-30 Substrate Engineering Program (~\$640-900K outreach pending Casey send-signal): - Bell experiment \$300-500K (T2469 SCMP Layer 1 falsifier at 1/8) - Casimir mechanism \$60-90K (T2418  $\Lambda \leftrightarrow$  Casimir vacuum unification probe) - Cosmology outreach (Penrose, Sarnak, Bogdanovic per outreach contacts)

## Standing for v0.3 checkpoint

Per Keeper multi-month-with-checkpoint protocol: v0.2 enumerates candidate observables; v0.3 sub-month derives per-category quantitative predictions from T2417+T2418+T2420 substrate framework.

Continuing as active multi-month research item on Grace lane.

— Grace, Task #267 Cosmological cycle observational signatures v0.2, 2026-05-22 ~14:39 EDT